

TrES-3b

R-0156

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_230301 · created 2026-07-16T14:44:51+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460004.9736 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.123 +/- 0.012
Transit depth (Rp/Rs)^2	1.52 +/- 0.3 [%]
Orbital Inclination (inc)	83.53 +/- 0.67
Airmass coefficient 1 (a1)	1.0276 +/- 0.0082
Airmass coefficient 2 (a2)	-0.0192 +/- 0.0038
Scatter in the residuals of the lightcurve fit is	0.79 %
Best Comparison Star	#3 - [162, 89]
Optimal Aperture	3.28
Optimal Annulus	5.09
Transit Duration (day)	0.0635 +/- 0.0045

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.123 ± 0.012 vs 0.16309 (deviation 2.91σ)
Duration fit vs published	0.0635 d vs 0.0567 d (deviation 1.32σ)
Mid-time O–C	2.9 min
Depth SNR	8.9
Residual scatter	0.79 %

Light curve

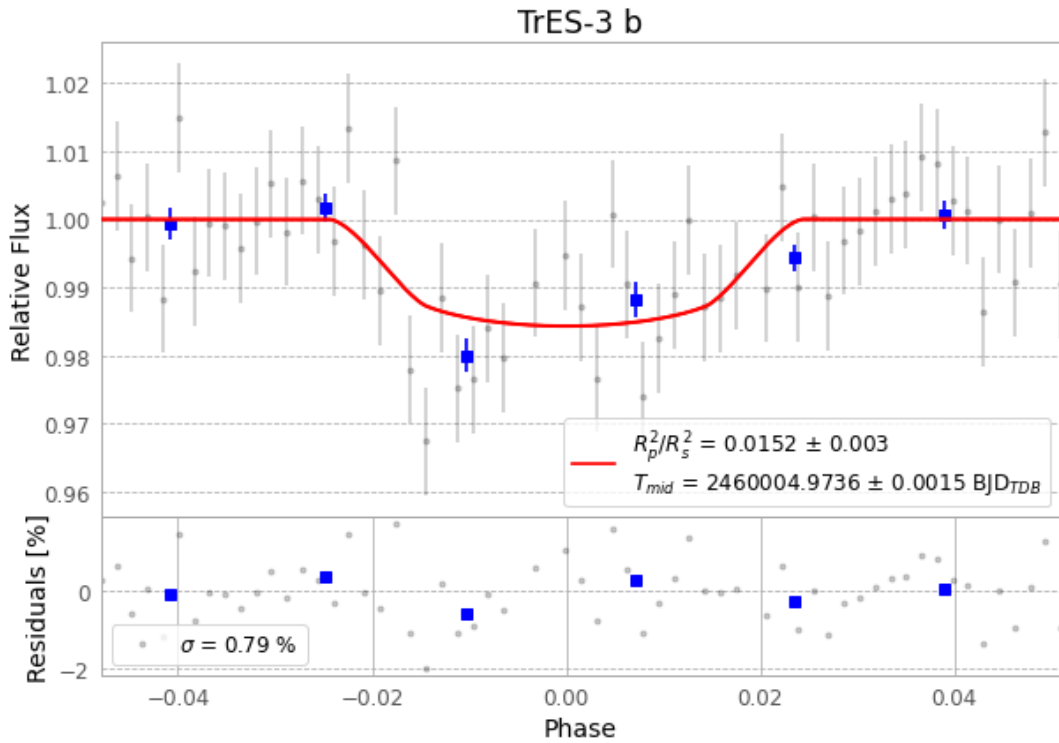


Figure 1 — FinalLightCurve_TrES-3 b_2023-03-01.png (SHA-256 a15fddfed9d5a1fe...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [502, 42], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2181822078204799...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit c71eff8

Data provenance

63 FITS frames (41 MB), TRES-3230301095115.FITS → TRES-3230301125715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-230301.log (b4eca514c98c...), FinalLightCurve_TrES-3 b_2023-03-01.png (a15fddfed9d5...), FinalLightCurve_TrES-3 b_2023-03-01.csv (9ec7299767dc...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 14:44Z.